

How to Measure a Lens **with Sag** on the LynceeTec Digital Holographer

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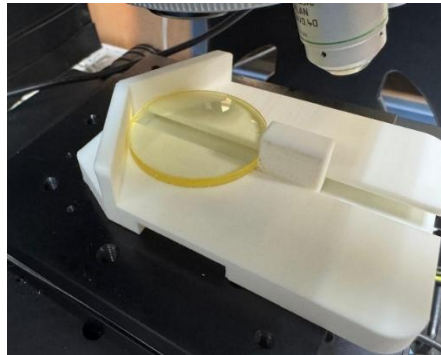
If you're very familiar with the LynceeTec and just want to see the improved options for measuring lenses with sag, skip to instruction [3e&c](#).

0. Measure Diameter and Minimum and Maximum Height of the Lens



1. Put the lens in the holder

The holder is designed for lenses with diameter $\leq 80mm$, minimum height $\geq 4mm$ and maximum height $\leq 9.5mm$.



Slide the slider until it presses against the end of the lens. If the holder isn't on the stage, the pegs on the holder insert right into the stage holes. You can rotate the stage by 90° to try and get more clearance if needed.

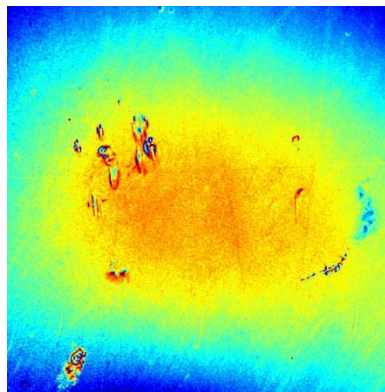
2. Setup Koala

- Launch Koala (username: "user" password: "user")
- Coarsely move the objective lens to the center of your object

- c) Change the lens to the 20x objective. Load the configuration, turn on the laser, open the phase diagram, and start automatic reconstruction.



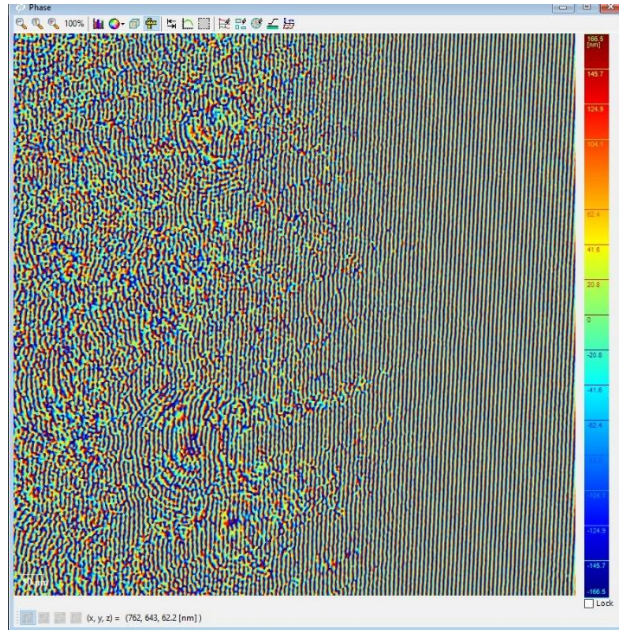
- d) Focus the microscope to the center of the lens. Aim for $z \approx 10\text{mm} - h$ (shown on the joystick) where h is height of the lens. It should look approximately like this



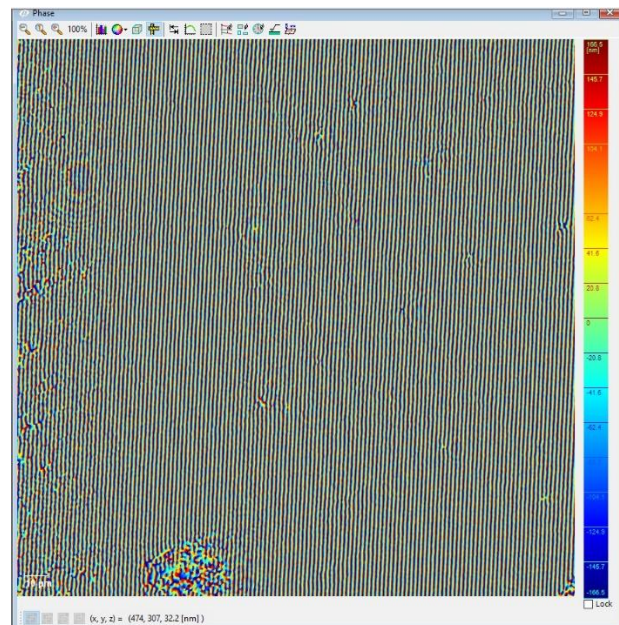
- e) If you only want to measure a profile from the center outwards, skip to step 3, if you want to measure over the full diameter, continue.
- f) On the joystick, make sure the XY axes are showing by pressing “Menu”, then zero the joystick by pressing “Rel”, “Zero”.
- g) Follow the lens to the edge by **moving the joystick to the right (screen goes to the left)¹**. Keep the surface in focus by sliding the right focus knob clockwise as you move toward the edge of the lens. You’ll know you hit the edge of the lens because no matter how you focus, the phase window stays as static. This is what the edge of a lens looks like²

¹ This is important because Koala can only stitch left to right.

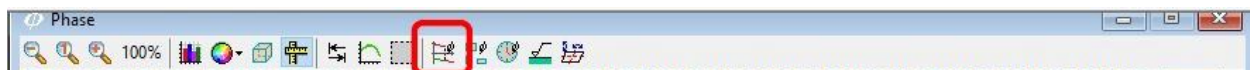
² A bit surprising to me. I’d expected it to look much straighter. Instead, the boundary is not clear.



- h) Move joystick to the left slightly (screen goes to the right) so that the entire phase window looks like nice vertical lines.



- i) Check the joystick's ΔY to make sure you didn't move in the Y direction. Move up and down until $\Delta Y = 0$.
- j) On the Phase window, enable unwrapping.

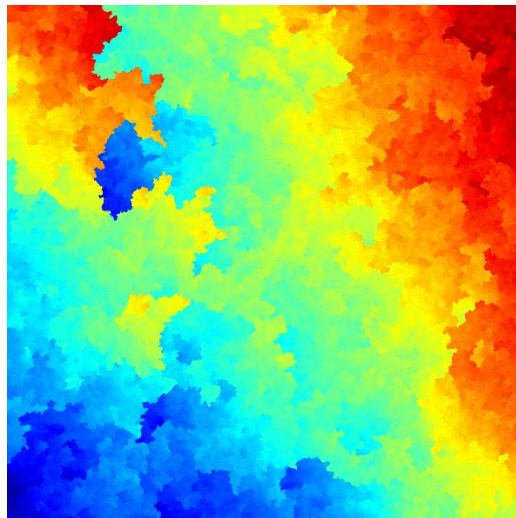


3. Stitching Settings

- a) Open TOOLS-*Working Distance* and Stitching windows

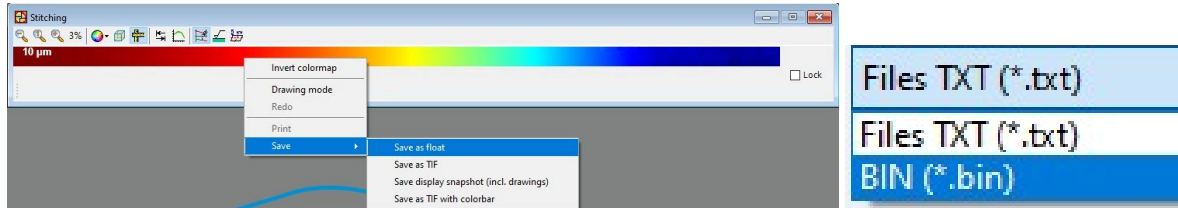


- b) Under “Stitching parameters” set X and Y overlap to 3, and uncertainty to 0.
- c) Under “Advanced settings” set min. contrast to a number between 75 and 85. For steeper objects, use higher percentage (at a tradeoff of time).
- d) Select “Stitch dimensions.” Set Y to 240. If you are measuring just the radius set X to the radius of your lens. If you want to measure the full diameter, set X to double ΔX shown on the joystick.
- e) Under Working Distance Search > Accurate Search, change Interval to 1500 and step to 25. If you’re working with a flatter object you can get away with up to 800 and 50. If 1500 and 25 loses the focus, try increasing it to 1800 20. Increasing Interval and decreasing step makes the scan more accurate at the cost of time.
- a. 1800 20 took me 2.5 hrs for a 50mm diameter lens, while 800 50 took 30 minutes for the same lens but it lost focus ~1mm from the end.
- f) Hit “Start Stitching”, “Yes” to the textbox and wait. If the Phase window starts looking like this, the microscope has lost the shape of the lens, and you need to restart with higher fidelity settings.



4. Analyzing the Shape

- a) Save the stitch by right-clicking the image and then Save > Save as Float. Make sure to switch the file type from .txt to .bin



- b) You now have a 2D array of heights on a grid which represents the surface of the profile you measured. To decode the file structure, see “C:\Israel\LynceeTec codes\read_mat_bin.m”. Feel free to analyze as you want.
- c) I wrote “C:\Israel\LynceeTec codes\mapProfile.m” which graphs the midline of the profile and fits it to circle and Bessel functions, and then calculates Bo , l_c , and R_{curv} .

